

How Many Householders Does State Tracking Need? An Exact Law for One-Layer Linear RNNs That Circuit Complexity Does Not Predict

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Abstract The usual account of why architectures fail at state tracking comes from circuit complexity. Log-precision Transformers can be simulated by constant-depth threshold circuits, and SSMs “cannot express computation outside the complexity class TC^0 ” [C-lit-merrill-tc0][C-lit-illusion]. We study one-layer linear RNNs whose token transitions are products of generalized Householder factors, as in DeltaNet and DeltaProduct [C-lit-deltaproduct][C-H-protocol]. Consider finite-state realisations of the word problem of a finite group G with alphabet Σ . A finite-state realisation with k factors per token exists if and only if $k \geq h^*(G, \Sigma)$, where h^* is a group invariant (proved by hand) [C-thm1]. The lower bound combines Lean-checked rank lemmas with a reviewed hand proof [C-L1][C-L4][C-L2]. For the non-solvable group A5 with involution inputs, $h^*(A5, \text{involutions}) = 1$ [C-T-A5-involutions]. With all letters, $h^*(A5, \text{all}) = 2$ [C-T-A5-all]. The S4 and A5 formats of prior work also need only 2 factors [C-T-S4-tn][C-T-A5-c3c5]. An atlas of all 319 groups of order at most 63 determines h^* exactly for 138 of them [C-atlas]. A preregistered training grid did not meet its success criteria [C-H-gates]. An exact verifier certifies every concrete instance [C-verifier][C-L3].

0.1 Introduction

The architecture debate on state tracking is usually framed in terms of circuit classes. Transformers with logarithmic arithmetic precision “can be simulated by constant-depth logspace-uniform threshold circuits” [C-lit-merrill-tc0]. SSMs are limited “very similarly to transformers” and “cannot solve simple state-tracking problems like permutation composition” [C-lit-illusion]. Self-attention “cannot model periodic finite-state languages, nor hierarchical structure, unless the number of layers or heads increases with input length” [C-lit-hahn]. Barrington’s result shows that bounded-width polynomial-size branching programs “recognize exactly those languages in NC1” [C-lit-barrington]. Empirically, Transformers can learn shortcuts to automata, and “ $O(1)$ -depth simulators are surprisingly common” [C-lit-liu]. LSTMs, by contrast, “can solve regular and counter-language tasks” [C-lit-deletang].

Linear RNNs with non-diagonal transitions have been proposed as a way out of this picture. Grazi et al. prove that positive-eigenvalue transitions cannot solve parity. They also show that “LRNNs can learn any regular language when their state-transition matrices are products of identity minus vector outer product matrices” [C-lit-grazzi]. DeltaProduct uses “products of n_h generalized Householder transformations”, which provides a tunable expressivity–efficiency trade-off [C-lit-deltaproduct]. RWKV-7 claims to “recognize all regular languages” [C-lit-rwkv7] and to solve the problem of “tracking swaps on five elements” [C-lit-rwkv7-swaps-body]. For diagonal SSMs, single-layer DCD models “cannot express state-

tracking of any non-Abelian group at finite precision” [C-lit-shakerinava].

These works leave a quantitative gap. The DeltaProduct authors report: “Unexpectedly, S4 and A5 can extrapolate robustly using only $n_h = 2$ despite the theorem suggesting 3 and 4, respectively” [C-lit-deltaproduct-body]. They attribute the efficiency to “isomorphism to subgroups of $SO(3, \mathbb{R})$ ” [C-lit-deltaproduct-so3-body]. The closest prior work states a representation law: “the minimal n_h that length-generalizes equals the maximal generator reflection length rank $(I - P)$ in the representation pinned by the task format (parity 1, S43, A5 / S54)” [C-lit-howe-law-body]. We ask which number of Householder factors per token is necessary and sufficient. Our answer does not depend on a chosen representation.

Contributions:

- We prove an exact law: a finite-state one-layer realisation with k Householder factors per token exists if and only if $k \geq h^*(G, \Sigma)$. The proof is by hand and is not machine-checked [C-thm1].
- We give a lower bound on Householder factors per token. It rests on two Lean-checked lemmas [C-L1][C-L4] and a compression lemma that is proved by hand and was reviewed by a red-team agent [C-L2].
- We certify instances. Non-solvable A5 with involution inputs needs exactly 1 factor [C-T-A5-involutions]. A5 with all letters needs exactly 2 [C-T-A5-all]. The S4 and A5 formats of prior work need exactly 2 [C-T-S4-tn][C-T-A5-c3c5].
- We build an atlas over all 319 groups of order at most 63 [C-atlas]. Our own character tables agree with GAP for every group [C-atlas].

- We ran a preregistered training grid. Its predictors and outcome are reported in full, including the learnability failures [C-H-protocol][C-H-gates].
- We describe an agentic lab with an exact verifier and a red team, and we document the loopholes it found and closed [C-lab][C-loophole1][C-redteam].

0.2 Setting and definitions

Word problem and one-layer realisation. Fix a finite group G and a generating alphabet Σ . We consider the one-layer recurrence $h_t = A(s_t) h_{t-1} + B(s_t)$ with an arbitrary readout [C-L2]. It *realises the word problem* of (G, Σ) if it outputs the group element of every prefix, for every length [C-L2]. The realisation is *finite-state* if only finitely many states are reachable [C-L2]. In the constructions, the readout maps $\rho(x) h_0$ to $\pi(x)$ [C-L3].

Transition families. We compare four families:

- `diag_pos`: diagonal transitions with entries in $[0, 1]$ [C-L5].
- `diag_pm`: real diagonal transitions with entries in $[-1, 1]$ [C-L5].
- `cdiag`: complex diagonal transitions [C-L5].
- Householder products: each token applies k generalized Householder factors $I - \beta u u^T$, i.e. DeltaNet or DeltaProduct with β in $[0, 2]$ [C-L1][C-H-protocol].

The invariants h and h^* . $h^*(G, \Sigma)$ is the least k with the following property [C-thm1]. There exist a finite group H , a surjection $\pi: H \rightarrow G$, generating lifts t_s of the letters, and a faithful real representation ρ of H , such that $\text{rank}(\rho(t_s) - I) \leq k$ for every letter. The invariant $h(G, \Sigma)$ is the same minimum restricted to $H = G$ [C-thm1]. h can be computed from the real character table [C-L6]. The fixed-space codimension is $\text{codim Fix } \rho(g) = \dim \rho - (1/|g|) \sum_j \chi_\rho(g^j)$ [C-L6]. Codimensions add over direct sums [C-L6]. A sum of real irreducible representations is faithful iff the kernels of the summands intersect trivially [C-L6].

0.3 Results I: theory

Lemma L1 (machine-checked). Suppose $R_1, \dots, R_k$ have rank at most one. Then $\text{rank}(\prod_i (I + R_i) - I) \leq k$ [C-L1]. In particular, a token transition made of k generalized Householder factors differs from the identity by rank at most k [C-L1]. The lemma is checked in Lean 4 with Mathlib as `rank_prod_sub_one_le` and `deltaproduct_rank_le` [C-L1]. As a negative control, Lean refutes the strengthened bound $k - 1$ (`strengthened_bound_false`) [C-L1]. All theorems use only the standard axioms `propext`, `Classical.choice` and `Quot.sound` [C-L1].

Lemma L4 (machine-checked). A real matrix of finite order with $\text{rank}(M - I) \leq 1$ is an involution [C-L4]. Hence, if ρ is a faithful real representation of a finite group H and $\pi(t) = s$ with s of order at least 3, then $\text{rank}(\rho(t) - I) > 1$ [C-L4]. It follows that $h^*(G, \Sigma) \geq 2$ whenever the alphabet contains a

letter of order at least 3. This holds even when a larger covering group is used [C-L4]. The Lean theorems are `sq_eq_one_of_rank_le_one_of_pow_eq_one` and `no_single_householder_lift` [C-L4].

Lemma L2 (compression lemma; proved by hand, not machine-checked). Suppose a one-layer recurrence with an arbitrary readout solves the word problem of (G, Σ) for every length, has finitely many reachable states, and satisfies $\text{rank}(A(s) - I) \leq k$ for every letter. Then $h^*(G, \Sigma) \leq k$ [C-L2].

The proof idea has three steps [C-L2]:

1. The reachable states form a finite transformation monoid that maps onto G .
2. An idempotent e of its minimal ideal gives a group eTe that maps onto G .
3. The compressed maps $A(u)A(s)$, restricted to the column space of the affine span of $e(Q)$, form a faithful representation of a covering group whose generators satisfy $\text{rank}(A(u)(A(s) - I)) \leq k$.

An independent red-team agent reviewed the proof. It found the proof valid for finite-state realisations, including affine input terms, β in $[0, 2]$, matrix-valued states and singular transitions [C-L2]. Lemmas L1 and L2 together give the lower bound for Householder products. A realisation with k factors per token has $\text{rank}(A(s) - I) \leq k$ [C-L1], so $k \geq h^*(G, \Sigma)$ [C-L2].

Lemma L3 (sufficiency; proved by hand, re-verified per instance). Suppose a finite group H maps onto G with generating lifts t_s , and ρ is a faithful real representation of H with $\text{rank}(\rho(t_s) - I) \leq k$. Then an exact finite-state one-layer realisation with k Householder reflections per token (β in $\{0, 2\}$) exists in dimension $\dim \rho$ [C-L3]. The construction has three steps [C-L3]:

1. An H -invariant inner product makes ρ orthogonal.
2. Cartan–Dieudonné factors each $\rho(t_s)$ into $\text{rank}(\rho(t_s) - I)$ reflections.
3. A generic initial state separates H .

The prior work cites the coarser form of Cartan–Dieudonné, that “every $n \times n$ orthogonal matrix can be written as a product of n reflections” [C-lit-grazzi-cd-body]. Our verifier re-checks Lemma L3 on every concrete instance [C-L3].

Theorem 1 (proved by hand, not machine-checked). Lemmas L2 and L3 together give the following statement [C-thm1]. For every finite group G and generating alphabet Σ , a finite-state one-layer realisation of the word problem with k Householder factors per token exists if and only if $k \geq h^*(G, \Sigma)$ [C-thm1]. In each instance, the parts of the proof have different status:

- The step from Householder factors to a rank bound is machine-checked [C-L1].
- The involution obstruction is machine-checked [C-L4].
- The compression step is a reviewed hand proof [C-L2].
- The constructions are certified exactly [C-L3].

Diagonal families (Lemma L5, proved by hand via the compression lemma). Finite-state one-layer realisations exist as follows [C-L5]:

- `diag_pos` (entries in $[0, 1]$): only for the trivial group.
- `diag_pm` (real entries in $[-1, 1]$): exactly for elementary abelian 2-groups.
- `cdiag` (complex entries): exactly for abelian groups.

This is consistent with prior theorems on diagonal SSMs [C-L5][C-lit-shakerinava][C-lit-grazzi].

Abelian groups (Lemma L7, proved by hand; instances certified). For an abelian group, $h^*(G, \Sigma)$ is 1 if every letter is an involution, and 2 otherwise [C-L7]. The upper bound uses a count cover that tracks each letter's count modulo its order [C-L7]. Its dimension is the number of involution letters plus twice the number of other letters [C-L7].

0.4 Results II: certified instances

Table 1 lists every certified instance with its lower bound, best certified realisation, faithful h and permutation-representation cost [C-T-Z2-all][C-T-A5-all][C-T-S5-all]. Each lower bound has one of two sources. It is either the trivial bound for a nontrivial group, or Lemma L4 (Lean-checked) combined with Lemma L2 (hand proof) [C-T-A5-all][C-T-A5-involutions].

A5 with involution inputs. Over faithful representations of A5 alone, $h(A5, \text{involutions}) = 2$ [C-T-A5-involutions]. Nevertheless, $h^*(A5, \text{involutions}) = 1$ [C-T-A5-involutions]. The certificate is the so3 construction with every involution letter lifted to its negative. Its covering group has order 120 and dimension 3 [C-T-A5-involutions]. A single reflection per token therefore tracks a non-solvable group exactly, provided the inputs are involutions and a covering group is used [C-T-A5-involutions].

A5 with all letters. $h^*(A5, \text{all}) = 2$ [C-T-A5-all]. The lower bound uses a letter of order at least 3 [C-T-A5-all]. The upper bound is the so3 construction with a covering group of order 60 in dimension 3 [C-T-A5-all]. The permutation representation costs 4 [C-T-A5-all]. The alphabets `cycles3` and `cycles5` also give $h^* = 2$ [C-T-A5-cycles3][C-T-A5-cycles5].

The generator formats of prior work. The closest prior work uses the formats “S 4 (transposition + 4-cycle), A 5 (3-cycle + 5-cycle; all-even), S 5 (transposition + 5-cycle; non-solvable)” [C-lit-howe-format-body]. We encode these as the two-letter alphabets `tn` and `c3c5` [C-T-S4-tn][C-T-A5-c3c5][C-T-S5-tn]. For S4, $h^*(S4, \text{tn}) = 2$ via so3 in dimension 3, while the permutation representation costs 3 [C-T-S4-tn]. For A5, $h^*(A5, \text{c3c5}) = 2$ via so3 in dimension 3, while the permutation representation costs 4 [C-T-A5-c3c5]. The permutation costs coincide with the values of the representation law, “S 4 3, A 5 / S 5 4” [C-lit-howe-law-body]. The exact minimum h^* is lower in both cases [C-T-S4-tn][C-T-A5-c3c5]. This explains the observation that “S 4 and A 5 can extrapolate robustly using only $n \cdot h = 2$ ” [C-lit-deltaproduct-body]. It is also consistent with the SO(3) explanation offered there [C-lit-

deltaproduct-so3-body]. With the full alphabet, S4 likewise has $h^*(S4, \text{all}) = 2$ [C-T-S4-all].

S5. For the format and for the full alphabet, the certified bounds are 2 and 4, so h^* is not determined [C-T-S5-tn][C-T-S5-all]. In both cases $h = 4$ over faithful representations [C-T-S5-tn][C-T-S5-all]. With transposition inputs, $h^*(S5, \text{transpositions}) = 1$ [C-T-S5-transpositions].

Abelian groups and Q8. For $Z2^3$, $h(Z2^3, \text{all}) = 3$ over faithful representations [C-T-Z2p3-all]. Nevertheless $h^*(Z2^3, \text{all}) = 1$, via a count cover of order 128 in dimension 7 [C-T-Z2p3-all][C-L7]. Similarly, $h(Z2^2, \text{all}) = 2$ while $h^*(Z2^2, \text{all}) = 1$ [C-T-Z2p2-all]. For Q8, $h(Q8, \text{all}) = 4$, and the certified bounds on h^* are 2 and 6 [C-T-Q8-all].

Contrast with circuit complexity. The solvable, abelian group Z3 requires $h^*(Z3, \text{all}) = 2$ [C-T-Z3-all]. The non-solvable group A5 with involution inputs requires $h^*(A5, \text{involutions}) = 1$ [C-T-A5-involutions]. Within one fixed group, the alphabet changes h^* : A5 has the values 1 and 2 [C-T-A5-involutions][C-T-A5-all], and S3 has the values 1 and 2 [C-T-S3-transpositions][C-T-S3-all]. The diagonal families separate differently. `diag_pm` realises Z2 but not Z3, while `cdiag` realises Z3 [C-T-Z2-all][C-T-Z3-all]. None of the three diagonal families realises S3 [C-T-S3-transpositions].

0.5 Results III: the atlas

The atlas covers all 319 groups of order at most 63 from the GAP SmallGroups library, each with the full alphabet [C-atlas]. Our own numerical character tables and GAP's exact tables give the same faithful h for every group, with 0 disagreements [C-atlas]. h^* is determined exactly for 138 of 319 groups [C-atlas]. These comprise 32 of the 213 non-abelian groups and every abelian group, the latter by Lemma L7 [C-atlas][C-L7]. The only non-solvable group in the atlas is A5. It has faithful $h = 2$ and $h^* = 2$, determined exactly [C-atlas-nonsolvable].

Faithful h is not monotone in solvability. 181 of the 212 solvable non-abelian groups have faithful $h \geq 3$ [C-atlas-monotone]. Every faithful representation of these groups therefore needs more Householder factors per token than the non-solvable A5 [C-atlas-monotone]. The largest faithful h in the atlas is 10, attained by C11:C5 [C-atlas-monotone]. For h^* , however, these groups only have certified bounds of the form $[2, \text{upper}]$, so we do not claim a strict separation in h^* [C-atlas-monotone]. Sign lifts to $G \times Z2$ multiply a letter's matrix by -1 [C-atlas-twist]. For 37 of the 213 non-abelian groups, they lower the certified upper bound below the faithful h [C-atlas-twist].

Figure 2 and Table 3 show the distribution of faithful h over the non-abelian groups [C-atlas-dist]. The counts are 2: 32, 3: 8, 4: 115, 5: 11, 6: 41, 8: 5, 10: 1 [C-atlas-dist].

0.6 Results IV: preregistered experiments

Protocol. The protocol (`prereg.md`) was committed before the first confirmatory run [C-H-protocol]. Its main parameters are [C-H-protocol]:

- Models. One-layer diag_pos, diag_pm, hh1, hh2, hh3 and hh4 (DeltaNet/DeltaProduct with beta in $[0, 2]$), plus an LSTM [C-H-protocol]. All use a 64-dimensional state and an MLP readout [C-H-protocol].
- Seeds and training. 20 seeds (1000-1019) per cell and 3000 steps, with a length curriculum from 8 to 64 and a final stage at 128 [C-H-protocol].
- Success criteria. A seed succeeds if its token accuracy on positions 257-512 of length-512 sequences is at least 0.9 [C-H-protocol]. A cell succeeds if at least 10 of 20 seeds succeed [C-H-protocol].
- Grid. 9 tasks x 7 architectures [C-H-protocol].

Figure 1 and Table 2 report the cell outcomes [C-H-protocol][C-T-A5-all].

Cells where our predictor was correct. In each of the following cells, our predictor said success and the cell succeeded:

- hh3 on Z2/all: 20 of 20 seeds succeed at 2x-4x the longest training length [C-G-hh3-Z2-all].
- hh4 on Z3/all: 12 of 20 [C-G-hh4-Z3-all].
- hh4 on S3/transpositions: 18 of 20 [C-G-hh4-S3-transpositions].
- hh4 on S3/all: 19 of 20 [C-G-hh4-S3-all].
- hh4 on S5/transpositions: 19 of 20 [C-G-hh4-S5-transpositions].
- hh4 on A5/involutions: 20 of 20 at 2x-4x, but 0 of 20 at 7x-8x [C-G-hh4-A5-involutions].
- hh3 and hh4 on Z2^3/all: 20 of 20 at 2x-4x [C-G-hh3-Z2p3-all][C-G-hh4-Z2p3-all], with 0 of 20 and 1 of 20 at 7x-8x [C-G-hh3-Z2p3-all][C-G-hh4-Z2p3-all].

Learnability failures. In each of the following cells, our predictor said success but the cell failed:

- hh4 on A5/all: 0 of 20 seeds succeed [C-G-hh4-A5-all]. Mean accuracy is 0.017 at chance 0.017, and in-distribution accuracy is 0.039 [C-G-hh4-A5-all]. The model did not fit even the training lengths, although hh4 exceeds $h^*(A5, \text{all}) = 2$ [C-T-A5-all].
- hh4 on S5/all: 0 of 20 seeds succeed, with in-distribution accuracy 0.024 [C-G-hh4-S5-all].
- hh3 on Z3/all: 6 of 20 seeds succeed [C-G-hh3-Z3-all].
- hh4 on Z2/all: 6 of 20 seeds succeed, with mean accuracy 0.515 at chance 0.500, even though its in-distribution accuracy is 1.000 [C-G-hh4-Z2-all].

The theory concerns the existence of exact solutions. These cells show that gradient training under this protocol does not reliably find such solutions [C-thm1][C-G-hh4-A5-all].

Predictor accuracy. On the determined cells, each predictor agrees with the cell outcomes as follows [C-H-accuracy]:

Predictor	Agreement
Our algebraic predictor	8 of 12 [C-H-accuracy]
Circuit class (solvable)	8 of 12 [C-H-accuracy]
Abelian	6 of 12 [C-H-accuracy]
Group size	8 of 12 [C-H-accuracy]
Representation law in the permutation representation	8 of 12 [C-H-accuracy]
Faithful representations only	8 of 12 [C-H-accuracy]

Our predictor therefore does not beat the baselines on this grid [C-H-accuracy].

Gates and controls. The preregistered criteria give the following results [C-H-gates]:

- Accuracy beats every baseline: False.
- All six discriminating tests significant after BH: False.
- Negative control clean: True.
- The overall hypothesis H-EX1: False.

The negative control NC-hh4-S5 trains on random targets in the cell hh4, S5, all [C-H-NC-hh4-S5]. It has successful seeds [0], $p = 1.000$, and a Benjamini-Hochberg adjusted $p = 1.000$ ($m = 1$, $q = 0.1$), so it is not significant [C-H-NC-hh4-S5]. The confirmatory experiment did not confirm our predictor. The exact law is a statement about expressivity, not about learnability [C-H-gates][C-thm1].

0.7 The agentic lab and the verification pipeline

The lab has six components [C-lab]:

- a literature scout with code-checked quotes;
- an integrator;
- a cascade of four researcher agents;
- an exact verifier;
- a red-team agent;
- a preregistration for every round.

The lab ran 9 rounds at a cost of 5.17 USD [C-lab]. 8 rounds produced an accepted claim, and all 8 survive re-checking with the hardened verifier (0 withdrawn) [C-lab]. 1 round produced a negative result [C-lab]. The scout kept 60 literature findings whose verbatim quote was found by code in the abstract [C-lab]. Before use, the verifier must pass a self-test of 46 known true and known false claims [C-verifier]. These include near-boundary cases, such as the same matrices without the sign lift and k one below the certified value [C-verifier].

Negative rounds and loopholes. The negative round asked whether one Householder reflection per token suffices for A5 with involution inputs. No claim from that round passed the verifier and the consistency rule [C-neg1]. Round 1 exposed a loophole [C-loophole1]. An agent answered ‘no’ and backed the answer with two true but weaker checks. The verifier, however, had refuted the agent’s claim $h^* = 2$ by finding a certificate with $k = 1$ itself [C-loophole1]. The consistency rule now requires an impossibility certificate for negative answers and exact-value checks for numbers [C-loophole1].

The agents’ own interpretations were recorded but not inspected. Several are inconsistent with the hardened rule.

In round 3 the agent interpreted $h^*(A5, \text{involutions})$ as “ > 1 ” [C-expressivity-R3-I], which contradicts the certified value 1 [C-T-A5-involutions]. In round 2 the interpretation “ $h^*(Z2^3, \text{all}) \leq 3$ ” is weaker than the certified value [C-expressivity-R2-I][C-T-Z2p3-all]. The red-team counter-check contested three rounds [C-expressivity-R4][C-expressivity-R6][C-expressivity-R8].

Red team. An independent red-team agent with read and execute rights only found three bugs [C-redteam]:

- The group names S1, D1 and D2 denoted wrong groups, so one false sentence each could be printed.
- The published h^* sentence omitted the finite-state hypothesis.
- The consistency rule accepted unchecked numbers in answer texts.

All three bugs were fixed and turned into regression cases [C-redteam]. The red team also confirmed the exact core against an independent re-implementation. 189 library constructions, 15556 exhaustive explicit-matrix claims and the character tables of all 319 SmallGroups were checked against GAP, with 0 disagreements [C-redteam].

0.8 Limitations and open questions

- Finite-state assumption. Lemma L2 and Theorem 1 assume finitely many reachable states [C-L2][C-thm1]. Realisations with unboundedly many states fall outside the law.
- Token-local transitions. In the recurrence $h_t = A(s_t) h_{t-1} + B(s_t)$, the transition depends only on the current letter [C-L2]. Architectures that let transitions depend on neighbouring tokens are not covered.
- Additive input pathway. The closest prior work reports that an additive input pathway acts as a “parasitic attractor”. Without that pathway, the same architecture “learns the exact automaton” [C-lit-howe]. Our compression lemma admits affine input terms for expressivity [C-L2], but our grid does not isolate this pathway [C-H-protocol].
- Exact versus finite-precision arithmetic. Our realisations are exact and certified in exact arithmetic [C-L3]. The cited impossibility results are stated at finite precision [C-lit-grazzi][C-lit-shakerinava].
- Hand proofs. The upper-bound half of the law rests on Lemma L3, a hand proof that is re-verified on every instance [C-L3]. Lemmas L2, L5 and L7 are hand proofs and are not machine-checked [C-L2][C-L5][C-L7].
- Open intervals. h^* remains open for S5/all, S5/tn and Q8/all [C-T-S5-all][C-T-S5-tn][C-T-Q8-all]. In the atlas, h^* is determined for only 138 of 319 groups [C-atlas].
- Multi-layer models. For diagonal SSMs, k layers suffice if and only if the group has “a subnormal series of length k , with Abelian factors” [C-lit-shakerinava]. A multi-layer analogue of h^* is open.
- Intermediate generation as a Householder budget. For Transformers, the gain from chain of thought “depends crucially on the amount of intermediate generation” [C-lit-cot]. With T steps of CoT, constant-depth Transformers can solve “any problem solvable by boolean circuits of size T ” [C-lit-li-cot]. Whether intermediate generation or padding acts as an additional Householder budget for one-layer linear RNNs is an open question.

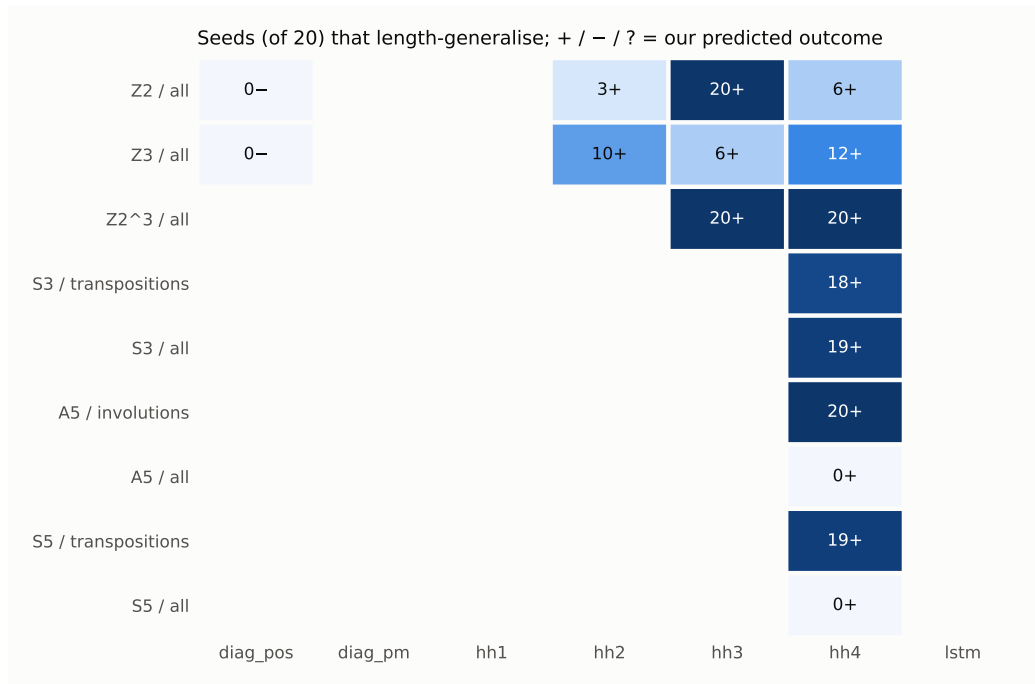


Figure 1: Preregistered grid: number of seeds (out of 20) whose accuracy on positions 257-512 is at least 0.9, with the outcome predicted by the algebraic law (+ success, - failure, ? undetermined). Table 2 is the table view. Claims [C-G-*].

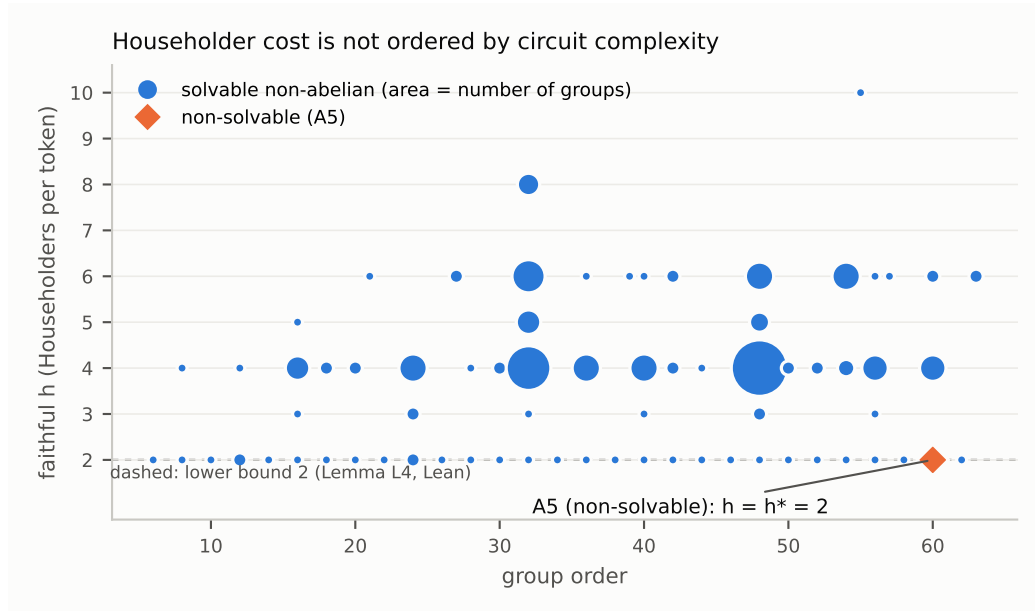


Figure 2: Atlas of all non-abelian groups of order at most 63: faithful h against group order (dot area = number of groups with that order and h). The only non-solvable group, A5, sits at the minimum value 2 while most solvable groups need more. Claims [C-atlas-monotone], [C-atlas-nonsolvable].

Table 1: Certified one-layer Householder complexity (exact verifier). lower: certified lower bound (Lemma L4 for 2); upper: best exact realisation found by the verifier (construction / sign twist, covering-group order $|H|$, dimension d); h: faithful representations of G only (own and GAP character tables); perm: cost in the permutation representation (the representation law of prior work). Claims [C-T*].

G	alphabet	order	solvable	lower	upper	construction	abs(H)	d	h	perm
Z2	all	2	yes	1	1	perm/none	2	2	1	1
Z3	all	3	yes	2	2	perm/none	3	3	2	2
Z5	all	5	yes	2	2	planar/none	5	2	2	4
Z6	all	6	yes	2	2	planar/none	6	2	2	5
Z2^2	all	4	yes	1	1	count/none	8	3	2	2
Z2^3	all	8	yes	1	1	count/none	128	7	3	3
S3	transpositions	6	yes	1	1	perm/none	6	3	1	1
S3	all	6	yes	2	2	perm/none	6	3	2	2
D4	all	8	yes	2	2	planar/none	8	2	2	3
D5	all	10	yes	2	2	planar/none	10	2	2	4
Q8	all	8	yes	2	6	perm/none	8	8	4	6
A4	all	12	yes	2	2	perm/none	12	4	2	2
S4	transpositions	24	yes	1	1	perm/none	24	4	1	1
S4	all	24	yes	2	2	so3/none	24	3	2	3
S4	tn	24	yes	2	2	so3/none	24	3	2	3
A5	c3c5	60	no	2	2	so3/none	60	3	2	4
S5	tn	120	no	2	4	perm/none	120	5	4	4
A5	involutions	60	no	1	1	so3/involutions	120	3	2	2
A5	cycles3	60	no	2	2	perm/none	60	5	2	2
A5	cycles5	60	no	2	2	so3/none	60	3	2	4
A5	all	60	no	2	2	so3/none	60	3	2	4
S5	transpositions	120	no	1	1	perm/none	120	5	1	1
S5	all	120	no	2	4	perm/none	120	5	4	4

Table 2: Preregistered grid: successful seeds out of 20 (accuracy ≥ 0.9 on positions 257-512 of length-512 sequences); mark: predicted success (+), predicted failure (-), undetermined (?) by our predictor. Claims [C-G*].

task	diag_pos	diag_pm	hh1	hh2	hh3	hh4	lstm
Z2/all	0 -	n/a	n/a	3 +	20 +	6 +	n/a
Z3/all	0 -	n/a	n/a	10 +	6 +	12 +	n/a
Z2^3/all	n/a	n/a	n/a	n/a	20 +	20 +	n/a
S3/transpositions	n/a	n/a	n/a	n/a	n/a	18 +	n/a
S3/all	n/a	n/a	n/a	n/a	n/a	19 +	n/a
A5/involutions	n/a	n/a	n/a	n/a	n/a	20 +	n/a
A5/all	n/a	n/a	n/a	n/a	n/a	0 +	n/a
S5/transpositions	n/a	n/a	n/a	n/a	n/a	19 +	n/a
S5/all	n/a	n/a	n/a	n/a	n/a	0 +	n/a

Table 3: Preregistered addendum H-EX2: the generator formats of the closest prior work. Successful seeds of 20 at 2x-4x (7x-8x); predicted outcome by our law ($h^* = 2$ for both tasks) and by the permutation-representation law. Claims [C-X*].

task	model	seeds 2x-4x	seeds 7x-8x	ours	permutation law
A5/c3c5	hh1	0	0	failure	failure
A5/c3c5	hh2	18	18	success	failure
A5/c3c5	hh3	18	10	success	failure
S4/tn	hh3	19	3	success	success

Table 4: Atlas, all 319 groups of order ≤ 63 , full alphabet. Claims [C-atlas*].

class	groups	h^* exact	faithful $h = 2$	faithful $h \geq 3$
abelian	105	105	62	42
solvable non-abelian	212	31	31	181
non-solvable	1	1	1	0

A Claim trace (Appendix C)

Every statement in the text cites claim identifiers in square brackets. The table lists every claim with its evidence level (proved_lean: Lean 4 + Mathlib; computed_rigorous: exact verifier; statistical: preregistered tests; observed; hypothesis: hand proof or uninspected interpretation) and the start of its canonical text. Full texts: [projects/expressivity/paper_evidence.json](#); proofs of the hand-proved lemmas: [expressivity/theory.md](#) (companion ledger).

claim	level	status	canonical text (start)
C-L1	proved_lean	confirmed	Lemma L1 (machine-checked in Lean 4 with Mathlib, theorems <code>rank_prod_sub_one_le</code> and <code>deltaproduct_rank_le</code>): if $R_1, \dots, R_k$ have rank at most one, the...
C-L4	proved_lean	confirmed	Lemma L4 (machine-checked in Lean 4 with Mathlib, theorems <code>sq_eq_one_of_rank_le_one_of_pow_eq_one</code> and <code>no_single_householder_lift</code>): a real matrix of fi...
C-L2	hypothesis	open	Lemma L2 (compression lemma; proved by hand, not machine-checked; reviewed by an independent red-team agent, which found it valid for finite-state rea...
C-L3	hypothesis	open	Lemma L3 (sufficiency; proved by hand, and re-verified on every concrete instance by the exact verifier): if a finite group H maps onto G with generat...
C-thm1	hypothesis	open	Theorem 1 (from Lemma L2, proved by hand, and Lemma L3): for every finite group G and generating alphabet Σ , a finite-state one-layer realisation ...
C-L5	hypothesis	open	Lemma L5 (proved by hand via the compression lemma, not machine-checked; consistent with prior theorems on diagonal SSMs): a finite-state one-layer re...
C-L7	hypothesis	open	Lemma L7 (proved by hand; instances certified): for an abelian group $h^*(G, \Sigma)$ is 1 if every letter is an involution and 2 otherwise; the upper bou...

C-L6	observed	confirmed	Computation of h (standard character theory): $\text{codim Fix } \rho(g) = \dim \rho - (1/g) \sum_j \chi_j \rho(g^j)$, codimensions add over direct sums, and a sum of...
C-T-Z2-all	computed_rigorous	confirmed	Certified (Z2, alphabet 'all', order 2, 1 letters, solvable): lower bound 1, best certified realisation $k = 1$ via perm/none (covering group of order 2...
C-T-Z3-all	computed_rigorous	confirmed	Certified (Z3, alphabet 'all', order 3, 2 letters, solvable): lower bound 2, best certified realisation $k = 2$ via perm/none (covering group of order 3...
C-T-Z5-all	computed_rigorous	confirmed	Certified (Z5, alphabet 'all', order 5, 4 letters, solvable): lower bound 2, best certified realisation $k = 2$ via planar/none (covering group of order...
C-T-Z6-all	computed_rigorous	confirmed	Certified (Z6, alphabet 'all', order 6, 5 letters, solvable): lower bound 2, best certified realisation $k = 2$ via planar/none (covering group of order...
C-T-Z2p2-all	computed_rigorous	confirmed	Certified ($Z2^2$, alphabet 'all', order 4, 3 letters, solvable): lower bound 1, best certified realisation $k = 1$ via count/none (covering group of orde...
C-T-Z2p3-all	computed_rigorous	confirmed	Certified ($Z2^3$, alphabet 'all', order 8, 7 letters, solvable): lower bound 1, best certified realisation $k = 1$ via count/none (covering group of orde...
C-T-S3-transpositions	computed_rigorous	confirmed	Certified (S3, alphabet 'transpositions', order 6, 3 letters, solvable): lower bound 1, best certified realisation $k = 1$ via perm/none (covering group...
C-T-S3-all	computed_rigorous	confirmed	Certified (S3, alphabet 'all', order 6, 5 letters, solvable): lower bound 2, best certified realisation $k = 2$ via perm/none (covering group of order 6...
C-T-D4-all	computed_rigorous	confirmed	Certified (D4, alphabet 'all', order 8, 7 letters, solvable): lower bound 2, best certified realisation $k = 2$ via planar/none (covering group of order...
C-T-D5-all	computed_rigorous	confirmed	Certified (D5, alphabet 'all', order 10, 9 letters, solvable): lower bound 2, best certified realisation $k = 2$ via planar/none (covering group of orde...
C-T-Q8-all	computed_rigorous	confirmed	Certified (Q8, alphabet 'all', order 8, 7 letters, solvable): lower bound 2, best certified realisation $k = 6$ via perm/none (covering group of order 8...
C-T-A4-all	computed_rigorous	confirmed	Certified (A4, alphabet 'all', order 12, 11 letters, solvable): lower bound 2, best certified realisation $k = 2$ via perm/none (covering group of order...
C-T-S4-transpositions	computed_rigorous	confirmed	Certified (S4, alphabet 'transpositions', order 24, 6 letters, solvable): lower bound 1, best certified realisation $k = 1$ via perm/none (covering grou...
C-T-S4-all	computed_rigorous	confirmed	Certified (S4, alphabet 'all', order 24, 23 letters, solvable): lower bound 2, best certified realisation $k = 2$ via so3/none (covering group of order ...
C-T-S4-tn	computed_rigorous	confirmed	Certified (S4, alphabet 'tn', order 24, 2 letters, solvable): lower bound 2, best certified realisation $k = 2$ via so3/none (covering group of order 24...
C-T-A5-c3c5	computed_rigorous	confirmed	Certified (A5, alphabet 'c3c5', order 60, 2 letters, non-solvable): lower bound 2, best certified realisation $k = 2$ via so3/none (covering group of or...
C-T-S5-tn	computed_rigorous	confirmed	Certified (S5, alphabet 'tn', order 120, 2 letters, non-solvable): lower bound 2, best certified realisation $k = 4$ via perm/none (covering group of or...
C-T-A5-involutions	computed_rigorous	confirmed	Certified (A5, alphabet 'involutions', order 60, 15 letters, non-solvable): lower bound 1, best certified realisation $k = 1$ via so3/involutions (cover...
C-T-A5-cycles3	computed_rigorous	confirmed	Certified (A5, alphabet 'cycles3', order 60, 20 letters, non-solvable): lower bound 2, best certified realisation $k = 2$ via perm/none (covering group ...
C-T-A5-cycles5	computed_rigorous	confirmed	Certified (A5, alphabet 'cycles5', order 60, 24 letters, non-solvable): lower bound 2, best certified realisation $k = 2$ via so3/none (covering group o...

C-T-A5-all	computed_rigorous	confirmed	Certified (A5, alphabet 'all', order 60, 59 letters, non-solvable): lower bound 2, best certified realisation k = 2 via so3/none (covering group of or...
C-T-S5-transpositions	computed_rigorous	confirmed	Certified (S5, alphabet 'transpositions', order 120, 10 letters, non-solvable): lower bound 1, best certified realisation k = 1 via perm/none (coverin...
C-T-S5-all	computed_rigorous	confirmed	Certified (S5, alphabet 'all', order 120, 119 letters, non-solvable): lower bound 2, best certified realisation k = 4 via perm/none (covering group of...
C-verifier	computed_rigorous	confirmed	The domain verifier passed its mandatory self-test of 46 known true and known false claims (near-boundary cases such as the same matrices without the ...
C-atlas	computed_rigorous	confirmed	Atlas over all 319 groups of order at most 63 (GAP SmallGroups library), full alphabet: own numerical character tables and GAP's exact tables give the...
C-atlas-nonsolvable	computed_rigorous	confirmed	In the atlas the only non-solvable group is A5 (order 60), with faithful h = 2 and h* = 2 determined exactly....
C-atlas-monotone	computed_rigorous	confirmed	Non-monotonicity: 181 of the 212 solvable non-abelian groups of order at most 63 have faithful h >= 3, i.e. every faithful representation of them need...
C-atlas-twist	computed_rigorous	confirmed	Sign lifts to G x Z2 (multiply a letter's matrix by -1) lower the certified upper bound below the faithful h for 37 of the 213 non-abelian groups....
C-atlas-dist	computed_rigorous	confirmed	Distribution of faithful h over the non-abelian groups of the atlas (value: number of groups): 2: 32, 3: 8, 4: 115, 5: 11, 6: 41, 8: 5, 10: 1....
C-H-protocol	observed	confirmed	Preregistered protocol (prereg.md, committed before the first confirmatory run): one-layer models diag_pos, diag_pm, hh1, hh2, hh3, hh4 (DeltaNet/Delt...
C-G-hh3-Z2-all	statistical	confirmed	Grid cell hh3 on Z2/all: 20 of 20 seeds succeed at 2x-4x the longest training length (18 of 20 at 7x-8x); mean accuracy 0.996 (chance 0.500); in-distr...
C-G-hh4-Z2-all	statistical	confirmed	Grid cell hh4 on Z2/all: 6 of 20 seeds succeed at 2x-4x the longest training length (7 of 20 at 7x-8x); mean accuracy 0.515 (chance 0.500); in-distrib...
C-G-hh3-Z3-all	statistical	confirmed	Grid cell hh3 on Z3/all: 6 of 20 seeds succeed at 2x-4x the longest training length (5 of 20 at 7x-8x); mean accuracy 0.444 (chance 0.333); in-distrib...
C-G-hh4-Z3-all	statistical	confirmed	Grid cell hh4 on Z3/all: 12 of 20 seeds succeed at 2x-4x the longest training length (10 of 20 at 7x-8x); mean accuracy 0.698 (chance 0.333); in-distr...
C-G-hh3-Z2p3-all	statistical	confirmed	Grid cell hh3 on Z2^3/all: 20 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.989 (chance 0.125); in-dist...
C-G-hh4-Z2p3-all	statistical	confirmed	Grid cell hh4 on Z2^3/all: 20 of 20 seeds succeed at 2x-4x the longest training length (1 of 20 at 7x-8x); mean accuracy 0.981 (chance 0.125); in-dist...
C-G-hh4-S3-transpositions	statistical	confirmed	Grid cell hh4 on S3/transpositions: 18 of 20 seeds succeed at 2x-4x the longest training length (12 of 20 at 7x-8x); mean accuracy 0.937 (chance 0.167...
C-G-hh4-S3-all	statistical	confirmed	Grid cell hh4 on S3/all: 19 of 20 seeds succeed at 2x-4x the longest training length (6 of 20 at 7x-8x); mean accuracy 0.971 (chance 0.167); in-distri...
C-G-hh4-A5-involutions	statistical	confirmed	Grid cell hh4 on A5/involutions: 20 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.990 (chance 0.017); i...
C-G-hh4-A5-all	statistical	confirmed	Grid cell hh4 on A5/all: 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.017 (chance 0.017); in-distrib...
C-G-hh4-S5-transpositions	statistical	confirmed	Grid cell hh4 on S5/transpositions: 19 of 20 seeds succeed at 2x-4x the longest training length (11 of 20 at 7x-8x); mean accuracy 0.964 (chance 0.008...

C-G-hh4-S5-all	statistical	confirmed	Grid cell hh4 on S5/all: 0 of 20 seeds succeed at 2x-4x the longest training length (0 of 20 at 7x-8x); mean accuracy 0.008 (chance 0.008); in-distrib...
C-H-accuracy	statistical	confirmed	Agreement of each predictor with the cell outcomes on the determined cells: our algebraic predictor 8 of 12; circuit class (solvable) 8 of 12; abelian...
C-H-NC-hh4-S5	statistical	confirmed	Preregistered test NC-hh4-S5 (negative control, random targets): cell ['hh4', 'S5', 'all'], successful seeds [0], $p = 1.000$, Benjamini-Hochberg adjust...
C-H-gates	statistical	confirmed	Preregistered success criteria: accuracy beats every baseline: False; all six discriminating tests significant after BH: False; negative control clean...
C-expressivity-R2	computed_rigorous	confirmed	Lab round 2. Question: For the abelian group Z_2^3 with alphabet 'all', compare $h(Z_2^3, \text{all})$ (faithful representations only) with $h^*(Z_2^3, \text{all})$. Which ...
C-expressivity-R2-I	hypothesis	open	Uninspected interpretation by the agent for expressivity-R2 (not a result; NOT consistent with the hardened consistency rule): $h(Z_2^3, \text{all}) = 3$ (faith...
C-expressivity-R3	computed_rigorous	confirmed	Lab round 3. Question: Liegt $h^*(A_5, \text{involutions})$ bei 1 oder darüber, wenn man nur ein einzelnes Householder-Produkt pro Token (HH_1) zulässt, und gilt...
C-expressivity-R3-I	hypothesis	open	Uninspected interpretation by the agent for expressivity-R3 (not a result; NOT consistent with the hardened consistency rule): $> 1...$
C-expressivity-R4	computed_rigorous	contested by red-team counter-check	Lab round 4. Question: For the quaternion group Q8 with alphabet 'all', what is $h(Q_8, \text{all})$, and how far apart are the certified bounds on $h^*(Q_8, \text{all})$?...
C-expressivity-R4-I	hypothesis	open	Uninspected interpretation by the agent for expressivity-R4 (not a result; NOT consistent with the hardened consistency rule): $h(Q_8, \text{all}) = 4...$
C-expressivity-R5	computed_rigorous	confirmed	Lab round 5. Question: Can a single layer with 2 Householder reflections per token track the non-solvable group A_5 with alphabet 'all' exactly? Determ...
C-expressivity-R5-I	hypothesis	open	Uninspected interpretation by the agent for expressivity-R5 (not a result; consistent with the hardened consistency rule): Ja, 2 Householder-Reflektio...
C-expressivity-R6	computed_rigorous	contested by red-team counter-check	Lab round 6. Question: Wie verhält sich $h(A_5, \text{involutions})$ gegenüber $h(A_5, \text{all})$, wenn man nur die Involutionen als Alphabet zulässt, und ist der Wer...
C-expressivity-R6-I	hypothesis	open	Uninspected interpretation by the agent for expressivity-R6 (not a result; NOT consistent with the hardened consistency rule): 2...
C-expressivity-R7	computed_rigorous	confirmed	Lab round 7. Question: For S4 with alphabet 'all', the permutation representation needs $\text{rank}(P(s) - I) = 3$ for 4-cycles. Is the true $h^*(S_4, \text{all})$ small...
C-expressivity-R7-I	hypothesis	open	Uninspected interpretation by the agent for expressivity-R7 (not a result; consistent with the hardened consistency rule): ja, $h^*(S_4, \text{all}) = 2...$
C-expressivity-R8	computed_rigorous	contested by red-team counter-check	Lab round 8. Question: Welche Gruppen mit Involutionen als Alphabet (z. B. S_4, D_4, Z_2^2 oder Q_8 mit geeignetem Twist) besitzen $h^*(G, \text{involutions}) = 1,...$
C-expressivity-R8-I	hypothesis	open	Uninspected interpretation by the agent for expressivity-R8 (not a result; consistent with the hardened consistency rule): $Z_2^2...$
C-expressivity-R9	computed_rigorous	confirmed	Lab round 9. Question: For S5 with alphabet 'all', what are the best certified lower and upper bounds on $h^*(S_5, \text{all})$, and what is $h(S_5, \text{all})$ over fait...
C-expressivity-R9-I	hypothesis	open	Uninspected interpretation by the agent for expressivity-R9 (not a result; NOT consistent with the hardened consistency rule): $h^*(S_5, \text{all})$: untere Schr...

C-neg1	observed	confirmed	Negative result of the lab (no claim passed the verifier and the consistency rule): [F3] Is one Householder reflection per token (plain DeltaNet with ...
C-lab	observed	confirmed	The agentic lab (literature scout with code-checked quotes, integrator, cascade of four researcher agents, exact verifier, red-team agent, per-round p...
C-loophole1	observed	confirmed	Loophole found in lab round 1 and closed: an agent answered 'no' (one reflection is not enough for A5 with involution inputs) and backed it with two t...
C-redteam	observed	confirmed	An independent red-team agent (read and execute only) found three bugs: group names S1, D1, D2 denoted wrong groups (one false sentence each could be ...
C-lit-merrill-tc0	observed	confirmed	Reference (arXiv:2207.00729, title verified by code against arXiv): "The Parallelism Tradeoff: Limitations of Log-Precision Transformers". Verbatim ab...
C-lit-illusion	observed	confirmed	Reference (arXiv:2404.08819, title verified by code against arXiv): "The Illusion of State in State-Space Models". Verbatim abstract quote checked by ...
C-lit-grazzi	observed	confirmed	Reference (arXiv:2411.12537, title verified by code against arXiv): "Unlocking State-Tracking in Linear RNNs Through Negative Eigenvalues". Verbatim a...
C-lit-deltaproduct	observed	confirmed	Reference (arXiv:2502.10297, title verified by code against arXiv): "DeltaProduct: Improving State-Tracking in Linear RNNs via Householder Products". ...
C-lit-rwkv7	observed	confirmed	Reference (arXiv:2503.14456, title verified by code against arXiv): "RWKV-7"Goose" with Expressive Dynamic State Evolution". Verbatim abstract quote ...
C-lit-sarrof	observed	confirmed	Reference (arXiv:2405.17394, title verified by code against arXiv): "The Expressive Capacity of State Space Models: A Formal Language Perspective". Ve...
C-lit-shakerinava	observed	confirmed	Reference (arXiv:2603.01959, title verified by code against arXiv): "The Expressive Limits of Diagonal SSMs for State-Tracking". Verbatim abstract quo...
C-lit-howe	observed	confirmed	Reference (arXiv:2609.18966, title verified by code against arXiv): "The Automaton Underneath: The Additive Input Pathway Is a Parasitic Attractor for...
C-lit-liu	observed	confirmed	Reference (arXiv:2210.10749, title verified by code against arXiv): "Transformers Learn Shortcuts to Automata". Verbatim abstract quote checked by cod...
C-lit-cot	observed	confirmed	Reference (arXiv:2310.07923, title verified by code against arXiv): "The Expressive Power of Transformers with Chain of Thought". Verbatim abstract qu...
C-lit-hahn	observed	confirmed	Reference (arXiv:1906.06755, title verified by code against arXiv): "Theoretical Limitations of Self-Attention in Neural Sequence Models". Verbatim ab...
C-lit-deletang	observed	confirmed	Reference (arXiv:2207.02098, title verified by code against arXiv): "Neural Networks and the Chomsky Hierarchy". Verbatim abstract quote checked by co...
C-lit-li-cot	observed	confirmed	Reference (arXiv:2402.12875, title verified by code against arXiv): "Chain of Thought Empowers Transformers to Solve Inherently Serial Problems". Verb...
C-lit-deltaproduct-body	observed	confirmed	Reference arXiv:2502.10297 (full-text quote checked by code against the arXiv HTML): "Unexpectedly, S 4 and A 5 can extrapolate robustly using only n ...

C-lit-deltaproduct-so3-body	observed	confirmed	Reference arXiv:2502.10297 (full-text quote checked by code against the arXiv HTML): "This efficiency arises from their isomorphism to subgroups of SO...
C-lit-howe-law-body	observed	confirmed	Reference arXiv:2609.18966 (full-text quote checked by code against the arXiv HTML): "the minimal n h that length-generalizes equals the maximal gener...
C-lit-howe-format-body	observed	confirmed	Reference arXiv:2609.18966 (full-text quote checked by code against the arXiv HTML): "S 4 (transposition + 4-cycle), A 5 (3-cycle + 5-cycle; all-even)...
C-lit-rwkv7-swaps-body	observed	confirmed	Reference arXiv:2503.14456 (full-text quote checked by code against the arXiv HTML): "RWKV-7 can, by Lemma 2 , solve the problem of tracking swaps on ...
C-lit-grazzi-cd-body	observed	confirmed	Reference arXiv:2411.12537 (full-text quote checked by code against the arXiv HTML): "every $n \times n$ orthogonal matrix can be written as a product of n r...
C-lit-barrington	observed	confirmed	Reference (doi:10.1016/0022-0000(89)90037-8, title verified by code against Crossref): "Bounded-width polynomial-size branching programs recognize exa...